

Module 9: Solana Virtual Machine (SVM)

Runtime Architecture and Program Execution

What is the SVM?

Solana Virtual Machine

- **Sealevel Runtime:** Solana's parallel smart contract execution engine
- **BPF Programs:** Berkeley Packet Filter bytecode for programs
- **Parallel Processing:** Execute multiple transactions simultaneously
- **Resource Management:** Compute units and gas metering

Key Differences from EVM

- **Parallel Execution:** Multiple transactions can run at once
- **Account-Based:** Programs and data stored in accounts
- **No Gas Wars:** Priority fees instead of gas auctions

SVM Architecture

Core Components

- **Programs:** Executable bytecode stored in accounts
- **Runtime:** Sealevel execution environment
- **Compute Units:** Resource allocation and metering
- **Parallel Processing:** Concurrent transaction execution

Execution Model

- **Single-threaded Accounts:** No race conditions
- **Cross-program Calls:** Programs can invoke other programs
- **State Consistency:** Atomic transaction execution

Architecture Overview



Executing a Program

```
// Execute single instruction
const execution = await svmService.executeProgram({
  programId: program.programId,
  instruction: {
    accounts: [/* account metas */],
    data: instructionData,
    programId: program.programId
  },
  signers: [payerKeypair],
  computeUnitLimit: 100000
});

console.log('Execution signature:', execution.transactionId);
console.log('Compute units used:', execution.computeUnitsUsed);
```

Parallel Execution

```
// Execute multiple instructions in parallel
const results = await svmService.executeParallel({
  executions: [
    { programId: prog1, instruction: instr1 },
    { programId: prog2, instruction: instr2 },
    { programId: prog3, instruction: instr3 }
  ],
  continueOnFailure: true, // Continue if one fails
  maxConcurrent: 5 // Limit concurrency
});

// Check results
results.forEach((result, index) => {
  if (result.status === 'fulfilled') {
    console.log(`Execution ${index} succeeded:`, result.value.transactionId);
  } else {
    console.log(`Execution ${index} failed:`, result.reason);
  }
});
```

Compute Units (CU) Management

CU Concepts

- **Compute Units:** Measure of computational work
- **CU Price:** Priority fee per CU
- **CU Limit:** Maximum allowed per transaction
- **CU Budget:** Total available for transaction

Setting CU Limits

```
// Set compute unit limit and price
const modifyComputeUnits = ComputeBudgetProgram.setComputeUnitLimit({
  units: 100000
});
```

```
const addPriorityFee = ComputeBudgetProgram.setComputeUnitPrice({
  microLamports: 5000 // 0.000005 SOL per CU
});
```

Metrics and Monitoring

Execution Analytics

- **Performance Tracking:** Execution time measurement
- **Resource Usage:** CU consumption monitoring
- **Success Rates:** Execution outcome statistics
- **Cost Analysis:** Gas expenditure tracking
- **Trend Analysis:** Historical performance patterns

Monitoring Endpoints

- `GET /svm/metrics/executions` - Get execution metrics
- `GET /svm/programs/:programId/stats` - Get program statistics
- `GET /svm/runtime/info` - Get runtime information

Security and Validation

Execution Security

- **Program Verification:** Bytecode validation before execution
- **Access Control:** Owner permission checks
- **Gas Limits:** Resource bound enforcement
- **Account Validation:** Address verification
- **Error Isolation:** Failure containment

Validation Checks

- Program ownership verification
- Account existence validation
- Instruction data sanitization
- Resource limit enforcement

Performance Optimization

Optimization Strategies

- **Parallel Execution:** Run independent operations concurrently
- **CU Optimization:** Right-size compute unit limits
- **Batch Processing:** Combine related operations
- **Caching:** Cache program and account data
- **Connection Pooling:** Efficient RPC connections

Scalability Features

- **Horizontal Scaling:** Multiple SVM service instances
- **Load Balancing:** Distribute execution requests
- **Async Processing:** Non-blocking execution
- **Resource Monitoring:** Track and limit resource usage

Integration with Solana

Web3.js Integration

- **PublicKey:** Address handling and validation
- **Transaction:** Transaction building and serialization
- **SystemProgram:** Account creation and management
- **ComputeBudgetProgram:** CU limit and priority setting
- **sendAndConfirmTransaction:** Transaction submission and confirmation

Network Support

- **Local Validator:** Development and testing
- **Devnet/Testnet:** Staging environments
- **Mainnet:** Production deployment
- **RPC Rotation:** Failover and load balancing

Next Steps

Module 9 Implementation Tasks

-  Program entity and CRUD operations
-  Program deployment functionality
-  Single program execution
-  Parallel transaction execution
-  Gas metering and tracking
-  SVM runtime monitoring
-  Local test validator setup
-  Multi-network support

Related Modules

SVM Runtime & Program Execution

- **Module 10:** Cross-Program Invocations

Thank You!

Questions?

SVM enables parallel smart contract execution on Solana!

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